

Introduction to Machine Learning

Study Notes

1. Why Do We Need Machine Learning?

Consider the problem of sorting emails into 'Spam' and 'Not Spam'. A genuine email might look like an interview invite with a legitimate meeting link, while a spam email typically promises prizes or money and asks for sensitive details.

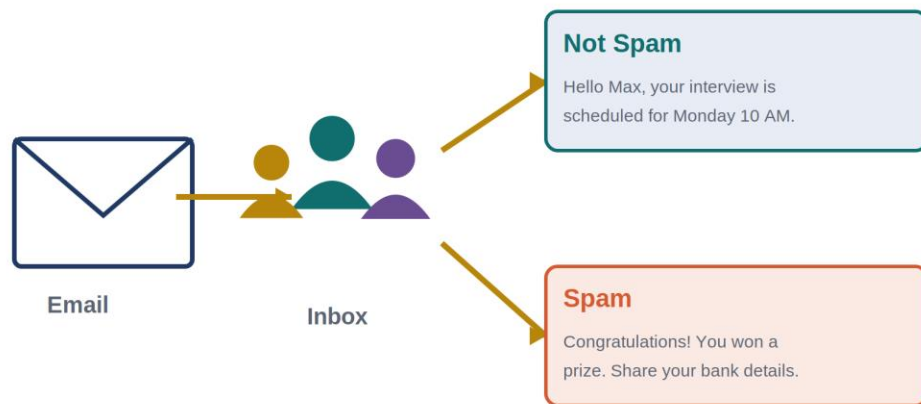


Figure 1: Every email needs to be routed to either the Spam or the Not Spam folder.

1.1 Approach 1 — Manual Checking

One option is to hire people to read and classify every incoming email manually. This does not scale well because of the following problems:

- Millions of emails arrive every day
- Hiring people for this is very expensive
- It takes too much time
- Reading personal emails raises privacy concerns



Hire people to read every single email

Figure 2: Manually reading every email does not scale.

1.2 Approach 2 — Rule-Based (If-Else) Systems

A second option is to hard-code rules, for example:

if "Congratulations" appears in the email → mark as Spam

if "Lottery" appears in the email → mark as Spam

This keyword list keeps growing (Lottery, Free Money, Prize, Gift, Reward, Bonus, and so on), which creates new problems:

- The rule set becomes huge and hard to maintain
- New spam keywords that aren't in the list are missed
- Rules need frequent manual updates

1.3 Approach 3 — Machine Learning

Instead of hand-writing rules, we give the computer a set of past emails that are already labelled as 'Spam' or 'Not Spam'. The computer learns the underlying pattern itself and can then classify brand-new, unseen emails.

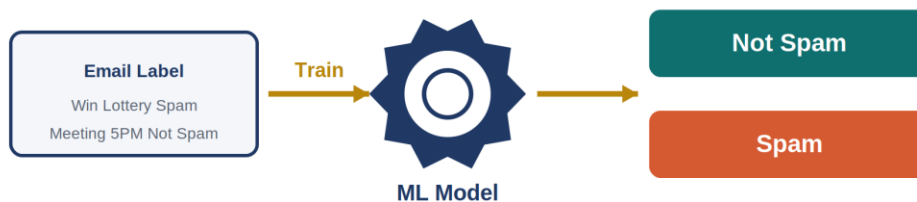


Figure 3: Labelled data is used to train an ML model, which then predicts labels for new emails.

Once trained, the model can look at a brand new email and predict whether it is spam — without anyone writing explicit rules.

2. What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence in which scientific algorithms learn patterns from past/present data, build a model, and use that model to make predictions about future or unseen data.



Figure 4: The general Machine Learning pipeline.

3. Real-Time Examples of Machine Learning



Figure 5: Five common real-world applications of Machine Learning.

3.1 Email Classification

Email providers such as Gmail and Outlook use ML models to automatically sort incoming mail into Spam or Not Spam.

3.2 Recommendation Systems

Streaming platforms (Netflix, Disney+ Hotstar, Prime Video) and e-commerce platforms (Flipkart, Myntra, Amazon) use ML to recommend movies, shows, or products based on user behaviour.

3.3 House Price Prediction

Real-estate platforms (Redfin, MagicBricks, Housing.com) use inputs such as area, number of bedrooms, and location to predict a house's price.

Input Features	Output
Sq.ft, No. of bedrooms, Location	Predicted House Price

3.4 Disease Detection

Using inputs like cold, fever, heart rate, and blood pressure, ML models can help flag the possible presence of a disease.

3.5 Loan Approval

Banks use factors such as income, occupation, and credit score to decide whether a loan application should be approved.

4. Types of Machine Learning

Machine Learning is broadly divided into four types:

1. Supervised Learning
2. Unsupervised Learning
3. Semi-supervised Learning
4. Reinforcement Learning

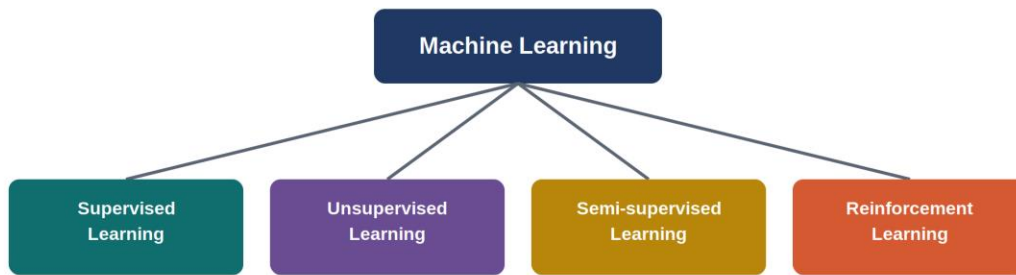


Figure 6: The four main types of Machine Learning.

4.1 Supervised Learning

Supervised Learning is the process of training a model by providing both the input and the correct output (label).

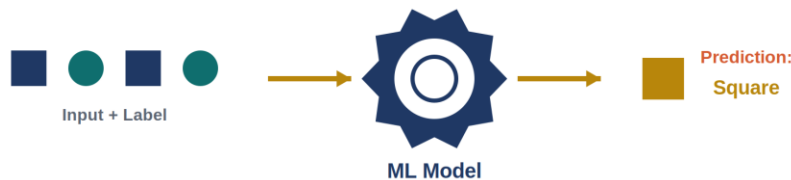


Figure 7: Supervised learning — the model is trained on labelled shapes and predicts the label of a new shape.

Example: given shapes labelled as 'Square' or 'Circle', the model learns to classify a brand-new shape correctly.

Supervised Learning is further divided into two types of problems:

- Classification Problem
- Regression Problem

4.1.1 Classification Problem

In a classification problem, the model predicts which category (class) the new input data belongs to. Example: classifying a fruit image as 'Apple' or 'Orange'.

Classification is further divided into:

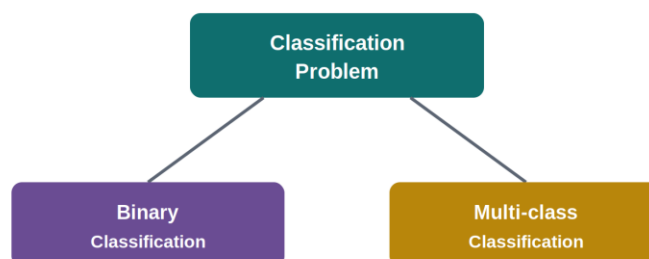


Figure 8: Binary vs. Multi-class classification.

Type	Description	Real-Time Examples
Binary Classification	Output column has exactly two categories	Email classification, Disease detection, Student result (Pass/Fail)
Multi-class Classification	Output column has more than two categories	Weather prediction, Student grade prediction, Movie genre classification

Sample binary classification data (student result prediction):

Marks	Attendance (%)	Assignment Submitted	Result
92	96	Yes	Pass
78	85	Yes	Pass
65	72	No	Fail
55	60	Yes	Fail
88	90	Yes	Pass

4.1.2 Regression Problem

In a regression problem, the output column contains continuous numeric values (rather than fixed categories).

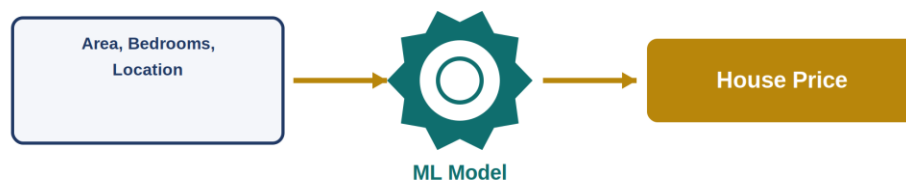


Figure 9: Regression — numeric inputs are mapped to a continuous numeric output (price).

Real-time examples:

- House price prediction
- Salary prediction
- Car price prediction

Area (sq.ft)	Bedrooms	Price (₹ Lakhs)
1000	2	45
1200	2	55
1500	3	72
1800	3	88
2200	4	115

4.2 Unsupervised Learning

Unsupervised Learning is the process of training a model by providing only input data, without any output (label).

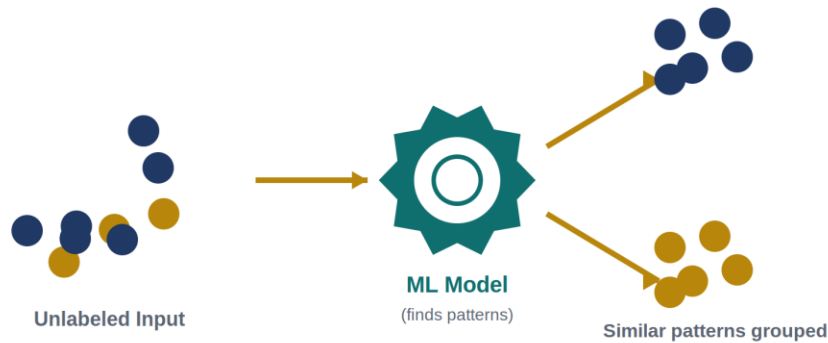


Figure 10: Unsupervised learning groups unlabelled data into clusters of similar points.

Unsupervised Learning is divided into:

- Clustering Problem
- Dimensionality Reduction

4.2.1 Clustering

Clustering is the process of grouping (segregating) data based on similar patterns, without any predefined labels.

Real-time examples:

- Customer segmentation
- News article segmentation

4.2.2 Dimensionality Reduction

Datasets can sometimes have a very large number of input columns (dimensions). This causes problems such as taking too much time to learn and producing a less efficient model.



Figure 11: Reducing the number of columns while keeping the important information.

Dimensionality Reduction is the process of reducing the number of columns in a dataset while preserving as much important information as possible. For example, columns like 'Owner', 'Owner Gender', and 'House Color' may be dropped because they don't meaningfully affect the house price prediction, while 'Sq.ft', 'Bedrooms', 'Bathrooms', and similar features can even be summarised into a single 'House size' category (Small / Medium / Large).

4.3 Semi-supervised Learning

When we have a small amount of labelled data and a large amount of unlabelled data, labelling all of it manually becomes costly. Semi-supervised Learning trains a model on the small labelled portion first, and then uses that model to label the remaining unlabelled data — effectively converting unlabelled data into labelled data.

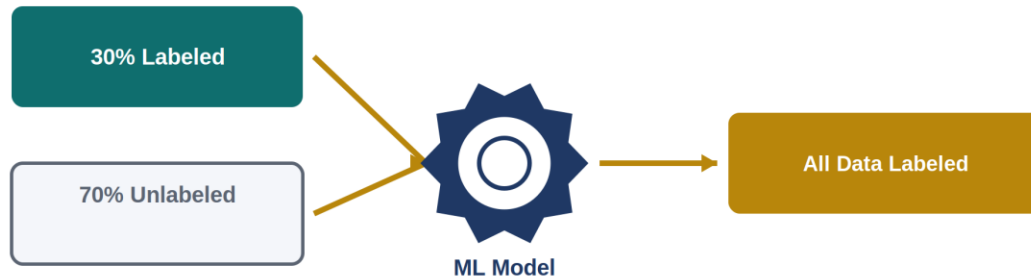


Figure 12: A small labelled subset is used to train a model that then labels the rest of the data.

4.4 Reinforcement Learning

Reinforcement Learning is a type of ML where an agent/model learns from experience through trial and error, guided by rewards and penalties.



Figure 13: The agent takes actions in an environment and learns from reward/penalty feedback.

Example — training a robotic arm to pick up objects and place them in a basket. The robot must learn:

- Where to move
- Which object to pick
- How to avoid obstacles
- How to reach the basket

Learning happens gradually, through repeated attempts:

Attempt	What Happened	Feedback	What the Robot Learns
1	Robot arm hits an obstacle (cone)	Penalty: -10	Hitting obstacles is bad
2	Robot picks up an object successfully	Reward: +10 (Penalty -10 still applies for earlier mistakes)	Picking an object gives a reward
3	Robot picks the object but drops it	Reward +10, then Penalty -10	Don't drop the objects

3 (improved)	Robot picks the object and places it in the basket correctly	Reward: +100 (highest so far)	The sequence of actions that avoids penalties and secures the placement gives the highest overall reward
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In short, Reinforcement Learning is built on the idea of trial and error combined with a reward-and-penalty feedback loop, allowing the agent to gradually discover the best sequence of actions.

5. Summary

- Machine Learning replaces manual work and rigid if-else rules by learning patterns directly from data.
- ML uses past/present data with algorithms to build a model that predicts future outcomes.
- Common real-world uses: email classification, recommendations, price prediction, disease detection, loan approval.
- Four types of ML: Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning.
- Supervised Learning → Classification (Binary / Multi-class) and Regression.
- Unsupervised Learning → Clustering and Dimensionality Reduction.
- Semi-supervised Learning is useful when labelled data is scarce and unlabelled data is abundant.
- Reinforcement Learning is based on trial-and-error with rewards and penalties.